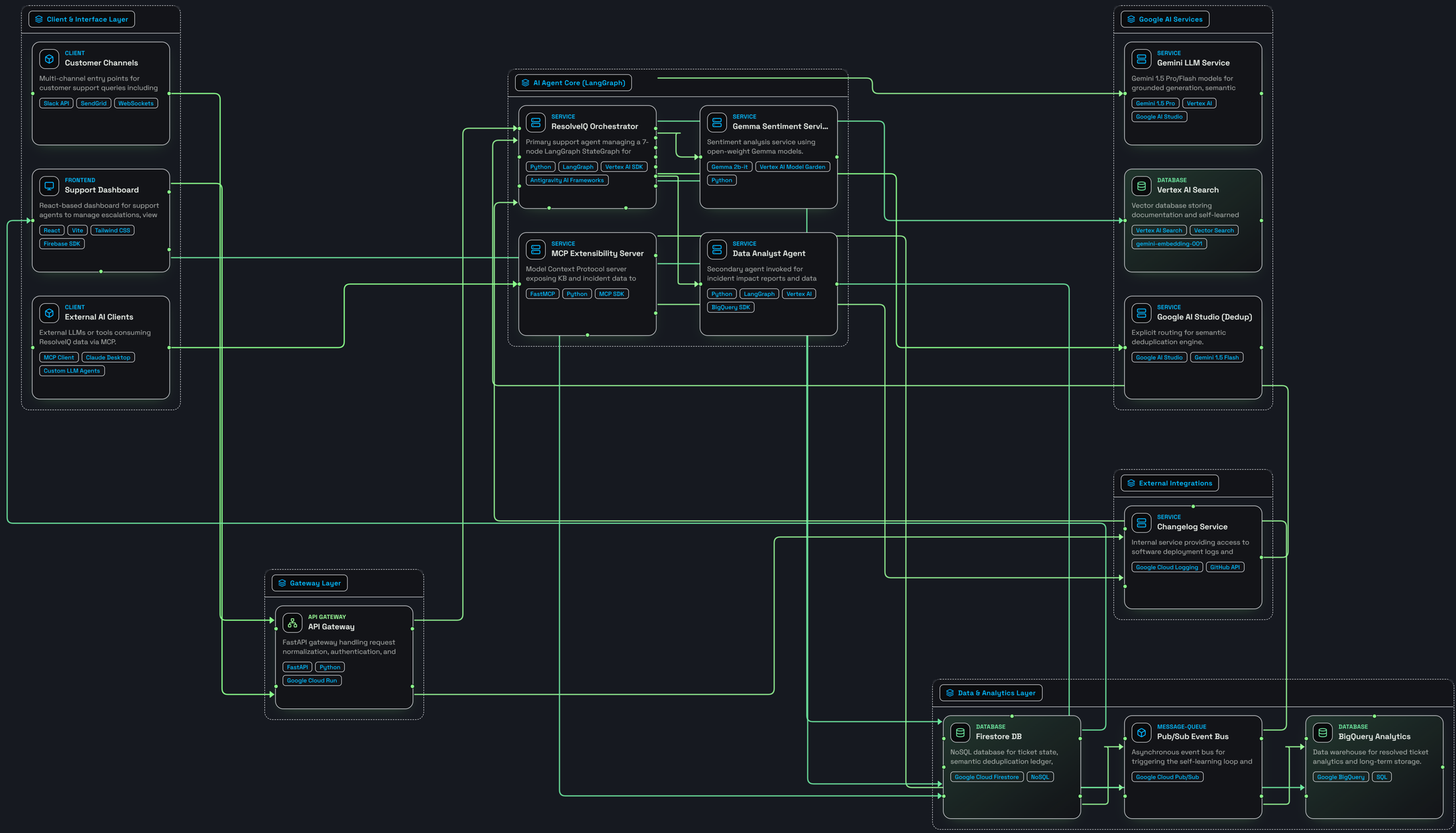
Client & Interface Layer
CLIENT
Customer Channels
Multi-channel entry points for customer support queries including
Slack API
SendGrid
WebSockets
FRONTEND
Support Dashboard
React-based dashboard for support agents to manage escalations, view
React
Vite
Tailwind CSS
Firebase SDK
CLIENT
External AI Clients
External LLMs or tools consuming ResolveIQ data via MCP.
MCP Client
Claude Desktop
Custom LLM Agents
AI Agent Core (LangGraph)
SERVICE
ResolveIQ Orchestrator
Primary support agent managing a 7-node LangGraph StateGraph for
Python
LangGraph
Vertex AI SDK
Antigravity AI Frameworks
SERVICE
Gemma Sentiment Servi...
Sentiment analysis service using open-weight Gemma models.
Gemma 2b-it
Vertex AI Model Garden
Python
SERVICE
MCP Extensibility Server
Model Context Protocol server exposing KB and incident data to
FastMCP
Python
MCP SDK
SERVICE
Data Analyst Agent
Secondary agent invoked for incident impact reports and data
Python
LangGraph
Vertex AI
BigQuery SDK
Google AI Services
SERVICE
Gemini LLM Service
Gemini 1.5 Pro/Flash models for grounded generation, semantic
Gemini 1.5 Pro
Vertex AI
Google AI Studio
DATABASE
Vertex AI Search
Vector database storing documentation and self-learned
Vertex AI Search
Vector Search
gemini-embedding-001
SERVICE
Google AI Studio (Dedup)
Explicit routing for semantic deduplication engine.
Google AI Studio
Gemini 1.5 Flash
External Integrations
SERVICE
Changelog Service
Internal service providing access to software deployment logs and
Google Cloud Logging
GitHub API
Gateway Layer
API GATEWAY
API Gateway
FastAPI gateway handling request normalization, authentication, and
FastAPI
Python
Google Cloud Run
Data & Analytics Layer
DATABASE
Firestore DB
NoSQL database for ticket state, semantic deduplication ledger,
Google Cloud Firestore
NoSQL
MESSAGE-QUEUE
Pub/Sub Event Bus
Asynchronous event bus for triggering the self-learning loop and
Google Cloud Pub/Sub
DATABASE
BigQuery Analytics
Data warehouse for resolved ticket analytics and long-term storage.
Google BigQuery
SQL